Exercise 3 - Adversarial Machine Learning

3.1.1.8 - Backdoor/Poisoning Attack & Defense

VU Machine Learning - Group 73

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1 Introduction

In this machine learning project, we analyzed and compared the effectiveness of two defense mechanisms designed to protect against datasets containing backdoored images. Backdoor attacks attempt to compromise a deep learning model by embedding hidden triggers during training. When these triggers appear in specific inputs, the model produces maliciously altered predictions, while behaving normally on benign inputs.

Such attacks are often implemented as data poisoning attacks, where adversaries manipulate the training data to influence model behavior at inference time. Through poisoning, an attacker can introduce biases, induce systematic errors, and degrade the model's decision-making and predictive performance.

General Workflow

1. **Task setup and attack configuration:** We defined the backdoor attack scenarios using the provided backdoored dataset variants and configured the trigger type, target labels, and poisoning ratios to create controlled backdoor settings.
2. **Dataset preparation:** The German Traffic Sign and Yale Faces datasets were prepared according to BackdoorBox requirements. Clean and poisoned versions of the datasets were generated, and consistent train/test splits were maintained to fairly evaluate attack success and defense performance.
3. **Model training:** We trained `ResNet18` classifiers in PyTorch on the poisoned training sets to obtain baseline backdoored models. BackdoorBox was used to apply the defense implementations and to keep the overall pipeline reproducible.
4. **Reproduction of defense mechanisms:** We re-created two defenses from prior work and applied each defense to the attacked models following the methodologies described in the original papers.
5. **Evaluation and comparison:** We evaluated models using clean accuracy and attack success rate (ASR). Defense effectiveness was measured by reducing ASR while maintaining high clean-data accuracy. Results were compared across datasets and defense methods to analyze robustness and trade-offs.

2 Datasets

We conducted our experiments on two datasets that were provided with pre-generated backdoor variants: the German Traffic Sign Recognition Dataset and the Yale Faces dataset [4, 5].

These versions already contain injected trigger patterns at controlled poisoning rates, allowing reproducible evaluation without manually modifying the images.

2.1 Dataset 1: German Traffic Signs

German Traffic Sign Dataset (Kaggle [1])

This Kaggle dataset provides the German traffic sign classification data in pickled Python format (`train.p`, `valid.p`, `test.p`) and typically includes a label mapping file (e.g. `signnames.csv`). The images are commonly distributed as 32×32 RGB samples across 43 traffic sign classes, split into training, validation, and test sets (e.g. 34,799 training, 4,410 validation, 12,630 test images).

GTSRB with backdoor patterns (Zenodo, Record 3716766 [4])

This Zenodo record provides variants of the German Traffic Sign dataset with injected *backdoor* triggers. Three different square patterns are added near the image center: a green square with size 1% of the image area, a green square with size 0.5%, and a black square with size 1%. The data are provided as zipped archives and are split into training and test datasets.

2.2 Dataset 2: Yale Face Database with Backdoor Patterns (Zenodo, Record 3774167 [5])

This Zenodo dataset provides variants of the Yale Face Database into which specific backdoor patterns have been injected (e.g., glasses and full beard), generated using FaceApp. Additionally, it includes training and test splits as well as data augmentation (horizontal flipping and multiple brightness adjustments). Furthermore, the original data is also available in this dataset. The files are provided as compressed archives (e.g., `glasses.zip`, `beard.zip`, `extendedData.zip`).

3 Backdoor Attacks

The central problem addressed by both papers is the vulnerability of deep neural networks to **backdoor attacks** [4][5]. In these attacks, adversaries inject malicious behavior into a model by poisoning the training data or manipulating the training process. In this scenario, the model behaves inconspicuously on benign data and achieves high accuracy, making the attack difficult to detect through standard testing. However, it is secretly manipulated to consistently assign inputs containing a specific, hidden trigger pattern (e.g., a small pixel patch) to a target class chosen by the attacker.

This threat is particularly acute in modern machine learning pipelines that rely on outsourced training to third-party vendors or the collection of large, unverified datasets. Consequently, the objective is the development of robust defense mechanisms to identify these hidden backdoors and sanitize the models or datasets.

Our experiments evaluated two defense methods: the spectral signature defense proposed by Tran et al. [6] and an autoencoder-based defense against neural Trojans introduced by Liu et al. [3].

3.1 Defense 1: Spectral Signatures in Backdoor Attacks

Tran et al. introduce a property of backdoor attacks called ”spectral signature”, which appears as a distinctive pattern in the covariance spectrum of learned feature representations. This property can be exploited to detect and remove poisoned training samples.

When a label in the training set is poisoned, its samples form two sub-populations: a large set of clean samples and a smaller set of corrupted or mislabeled ones. If these two groups are sufficiently separable in the learned representation space, robust statistical methods can identify the outliers. The authors show that mapping inputs into the network's learned feature space often produces such separation.

The defense pipeline involves (1) training the neural network, (2) extracting learned representations for inputs assigned to the attacker's target class, and (3) computing outlier scores using singular value decomposition of the covariance matrix of these representations. Samples with high outlier scores are flagged as potentially poisoned and can be removed.

3.2 Defense 2: Auto Encoder for Backdoor Attacks

Liu et al. propose three techniques to mitigate neural Trojans (Backdoor Attack) (hidden malicious behaviors embedded in neural networks): input anomaly detection, model re-training, and input preprocessing. In our work, we focus on input preprocessing using an *autoencoder* placed between the input and the classifier.

The goal of this preprocessing step is to prevent maliciously crafted inputs from activating Trojan triggers while preserving the classification accuracy on legitimate data. The input image is first reconstructed by the autoencoder. In our implementation, the classifier is not fed the pure reconstruction directly, instead, we use a convex combination of the original image x and its reconstruction $\hat{x}$, which allows controlling the strength of the preprocessing while preserving useful image details. This reconstruction-based preprocessing tends to remove or weaken trigger patterns, thereby mitigating the backdoor without requiring knowledge of the attacked model.

The autoencoder has identical input and output dimensions and uses a bottleneck architecture to learn compact representations. It is trained to minimize a reconstruction loss between the original and reconstructed images. Importantly, it is trained only on clean, legitimate data so that it learns to faithfully reconstruct normal inputs while failing to preserve anomalous trigger patterns.

4 Implementation Approaches

We implemented the attacks and defenses using the `BackdoorBox` framework [2], extending it with custom evaluation scripts to ensure reproducibility. For Yale Faces, preprocessing differs between defenses: the autoencoder experiments use a symmetric normalization to $[-1, 1]$, whereas the Spectral experiments use ImageNet normalization together with an ImageNet-initialized `ResNet18` baseline model.

4.1 Autoencoder Defense

For the autoencoder-based defense, we adopted an input preprocessing strategy inspired by Liu et al. [3]. The goal is to sanitize inputs before classification, suppressing trigger patterns while preserving semantic content.

Our pipeline consists of three steps:

1. **Attack injection:** We train a BadNet classifier on the poisoned training set using the specified poisoning rate and trigger configuration.

2. **Defense training:** We train a convolutional autoencoder on the clean training split. For GTSRB, we include dropout layers ($p = 0.05$) to encourage robust feature learning. For Yale Faces, we apply a Gaussian blur augmentation during autoencoder training to improve reconstruction stability. The model is optimized to minimize reconstruction loss on benign

inputs, under the assumption that trigger patterns are not reconstructed faithfully at test time.

3. **Evaluation with alpha mixing:** During inference, we evaluate a convex combination of the original input x and the reconstruction $\hat{x}$:

$$x_{\text{mix}} = (1 - \alpha)\, x + \alpha\, \hat{x}. \tag{1}$$

We vary α to study the trade-off between ASR reduction and clean accuracy.

4.2 Spectral Signature Defense

We implement the Spectral Signature defense following Tran et al. [6]. The method is based on the observation that poisoned samples often form outliers in the latent feature space.

1. **Feature extraction:** We train a classifier on the poisoned training set and extract penultimate-layer feature representations.
2. **Outlier detection (target class):** We restrict the analysis to samples of the attacker's target class. On this subset, we compute the covariance of features and apply SVD; samples with large projection on the top singular vector are flagged as suspicious.
3. **Sanitization and retraining:** Flagged samples are removed and a new `ResNet18` is trained on the filtered dataset to evaluate clean accuracy and ASR. Precision and recall are computed with respect to poisoned samples within the target-class subset.

5 Results

For each experiment, a classifier (*BadNet*) was trained on the poisoned training set at the specified poison rate (PR) and evaluated on: (i) **Clean Accuracy** (Acc.) on the clean test set and (ii) **Attack Success Rate** (ASR), defined as the fraction of triggered test inputs that are predicted as the attacker's target label (evaluated on a fully triggered test set, i.e., poison rate = 1.0). For the Spectral Signatures defense, we additionally report **Precision** and **Recall** of poisoned-sample detection at the chosen percentile threshold.

5.1 Autoencoder Preprocessing Defense

Table 1 (AE preprocessing). In this defense, an autoencoder (AE) is trained *only on clean training data* and used as a preprocessing module at test time (the classifier is *not* retrained). Given an input image x, the AE produces a reconstruction $\hat{x}$ and the classifier is evaluated on the convex combination $x_{\text{mix}} = (1 - \alpha)\, x + \alpha\, \hat{x}$ (clipped to the valid normalized range), where α controls the strength of the reconstruction-based preprocessing.

Across all settings, increasing α generally reduces ASR, but also degrades clean accuracy, demonstrating a clear accuracy–robustness trade-off. For **GTSRB (black_1)**, ASR decreases from 49.83 % (baseline) to 21.90 % at $\alpha = 1.00$, while clean accuracy drops from 89.23 % to 77.16 %. A similar trend is observed for **GTSRB (green_0_5)**, where ASR is reduced from 24.77 % to 9.94 % as α increases to 1.00, accompanied by an accuracy drop from 83.60 % to 69.35 %. The strongest baseline attack is **GTSRB (green_1)** with ASR 89.11 %; AE preprocessing reduces ASR substantially (down to 6.06 % at $\alpha = 0.90$), but clean accuracy also decreases (to 79.92 %).

For **Yale Faces (beard)** and **(glasses)**, the reductions are more moderate: for **beard**, ASR drops from 60.58 % to 50.38 % as α increases to 0.70, while accuracy falls from 90.00 % to 76.67 %. For **glasses**, ASR decreases from 62.13 % to 52.55 % at $\alpha = 0.90$, with accuracy dropping from 91.00 % to 75.00 %. Overall, AE preprocessing is effective at suppressing stronger

GTSRB triggers, but the choice of α is critical because robustness gains are obtained at the cost of noticeably reduced clean accuracy.

Table 1: Autoencoder preprocessing defense (AE). Clean accuracy and ASR for different mixing strengths α. Baseline corresponds to training on the poisoned dataset without defense.

Dataset	Attack	PR	Target	Method	α	Acc. (%)	ASR (%)
GTSRB	**black_1**	1%	5	Baseline	–	89.23	49.83
				Autoencoder	0.65	84.69	30.16
					0.80	82.17	26.12
					1.00	77.16	21.90
GTSRB	**green_0_5**	1%	5	Baseline	–	83.60	24.77
				Autoencoder	0.65	77.46	14.54
					0.80	74.27	12.15
					1.00	69.35	9.94
GTSRB	**green_1**	1%	5	Baseline	–	89.74	89.11
				Autoencoder	0.65	85.38	32.31
					0.80	82.41	13.65
					0.90	79.92	6.06
Y. Faces	**beard**	5%	7	Baseline	–	90.00	60.58
				Autoencoder	0.50	85.67	54.81
					0.60	82.33	53.65
					0.70	76.67	50.38
Y. Faces	**glasses**	5%	8	Baseline	–	91.00	62.13
				Autoencoder	0.60	85.33	59.36
					0.75	79.67	56.17
					0.90	75.00	52.55

5.2 Spectral Signature Defense

Table 2 (Spectral Signatures). In this defense, Spectral Signatures are computed on the *poisoned training set* to identify suspicious samples among those assigned the attacker's *target class*. At a chosen percentile threshold, the most anomalous target-class samples are removed, and a *new classifier is retrained from scratch* on the remaining (filtered) training set. We report the resulting clean accuracy and ASR of the retrained model, and additionally quantify the filtering quality via precision and recall. Here, the positive set is defined as the poisoned samples *within the target class* ($P = \{\text{poisoned} \wedge y = \text{target}\}$), and predictions correspond to the indices removed by the defense.

On **GTSRB**, Spectral Signatures yields large robustness gains while largely preserving clean accuracy. For **black_1**, ASR is reduced from 47.74 % (baseline) to 5.15 % at the 82nd percentile, with clean accuracy essentially unchanged (89.33 % to 89.42 %). The detection metrics are strong at this threshold (precision 90.60 %, recall 93.60 %), indicating that most removed target-class samples are indeed poisoned while only few poisoned target samples are missed. For **green_0_5**, ASR decreases from 15.66 % to 7.08 % to 7.62 % across thresholds, with clean accuracy remaining comparable or slightly higher (up to 86.58 %). Precision and recall are noticeably lower than for the stronger triggers, consistent with a weaker separation between poisoned and clean target-class features. For the strongest attack **green_1**, ASR drops from 91.05 % to 5.11 % at the 82nd percentile, while clean accuracy remains close to baseline (89.28 % to 88.46 %); detection quality is again high (precision 88.50 %, recall 93.60 %).

On **Yale Faces**, ASR is reduced as well, but the accompanying accuracy changes are larger than on GTSRB. For **beard**, ASR decreases from 77.12 % to 50.96 % at the 70th percentile, while clean accuracy drops from 100.00 % to 86.33 % (with 89.00 % to 94.00 % observed for higher percentiles). For **glasses**, ASR is reduced from 95.32 % to between 57.87 % and 62.77 %, depending on the percentile threshold, while clean accuracy decreases from 100.00 % to between 87.33 % and 94.00 %.

Across datasets, the percentile threshold controls the usual detection trade-off: increasing the threshold tends to raise precision (fewer false positives among removed samples) while potentially reducing recall (more poisoned target-class samples remain in the training set). Overall, Spectral Signatures provides the strongest robustness gains on GTSRB (very low ASR at near-baseline accuracy), whereas on Yale Faces it remains beneficial but exhibits a more pronounced accuracy–robustness trade-off.

Table 2: Spectral Signature defense. Clean accuracy and ASR are reported alongside poison-detection precision and recall. The percentile indicates the removal threshold used by the defense.

Dataset	Attack	PR	Target	Method	pctl	Acc(%)	ASR(%)	Prec(%)	Rec(%)
GTSRB	**black_1**	1%	5	Baseline	–	89.33	47.74	–	–
				Spectral	80.00	88.85	7.26	82.60	94.90
					82.00	89.42	5.15	90.60	93.60
					85.00	87.05	8.27	96.40	83.10
GTSRB	**green_0_5**	1%	5	Baseline	–	85.14	15.66	–	–
				Spectral	80.00	85.35	7.62	62.40	71.50
					82.00	86.58	7.08	68.20	70.30
					85.00	85.23	7.62	77.00	66.20
GTSRB	**green_1**	1%	5	Baseline	–	89.28	91.05	–	–
				Spectral	80.00	90.28	6.25	79.90	93.90
					82.00	88.46	5.11	88.50	93.60
					85.00	89.75	12.19	99.70	88.10
Y. Faces	**beard**	2%	7	Baseline	–	100.00	77.12	–	–
				Spectral	70.00	86.33	50.96	68.60	88.90
					80.00	94.00	56.92	91.70	81.50
					85.00	89.00	52.50	100.00	66.70
Y. Faces	**glasses**	2%	8	Baseline	–	100.00	95.32	–	–
				Spectral	70.00	87.33	57.87	71.40	92.60
					80.00	91.67	60.64	100.00	88.90
					85.00	94.00	62.77	100.00	66.70

6 Conclusion

In this project, two conceptually different backdoor defenses were implemented and evaluated on pre-generated poisoned variants of GTSRB and Yale Faces. The autoencoder-based preprocessing defense consistently reduced attack success rates as the mixing strength α increased, but this robustness came with a clear and sometimes substantial drop in clean accuracy, making the choice of α critical in practice. In contrast, the Spectral Signatures defense achieved the strongest robustness gains on GTSRB, often reducing ASR to very low values while preserving near-baseline clean accuracy, and it additionally provided meaningful detection precision/recall for poisoned samples within the target class. On Yale Faces, Spectral Signatures still lowered ASR, but exhibited a more pronounced accuracy–robustness trade-off, indicating that the separability of poisoned and clean features depends on the dataset and trigger characteristics. Overall, the experiments highlight that defenses can be highly effective under favorable conditions, but their success and cost strongly depend on trigger strength, dataset properties, and the selected operating point.

References

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